

Capstone — From Data to Decision

In groups of **3–4**, build one small end-to-end data product in **3 hours**. Each member owns a role (a slice of an earlier activity); everyone feeds **one shared Google Sheet**, then you combine and present.

Roles (work in parallel)

1 • Collector

slice of Day 1 Activity 1 / Day 2 Activity 3

- Create a short **Google Form** (3–4 questions) → link a response Sheet. Group fills it for a few rows.
- (Or) run the IoT simulator so it writes to a Sheet.

2 • Scraper / Cleaner

slice of Day 1 Activity 2

- Scrape your assigned practice portal: `tables = pd.read_html(URL)`.
- Clean: `df = df.drop_duplicates()`, fill missing numbers with the mean, missing text with the mode.
- Check one correlation: `df['A'].corr(df['B'])`.

3 • Analyst

slice of Day 1 Activity 3

- Open the **generic EDA notebook**, set the config labels, run group-by + dashboard.
- Answer the **3 questions**: top/bottom group, the gap, and one relationship.

4 • AI / Builder

slice of Day 2 Activity 2 / 3 / 1

- **Classify** a few images with a reused Teachable Machine model (upload the zip — it auto-arranges), **and/or**
- Build one **AppSheet** screen on the Sheet, or one **n8n** email alert.

3-hour timetable

Time	What happens
0:00–0:15	Brief, form groups, assign roles + your practice portal
0:15–1:15	Parallel build (60 min) — all roles at once
1:15–2:00	Integrate (45) — notebook reads the shared Sheet; clean → correlation → dashboard → 3 questions; Builder connects app/alert
2:00–2:40	Polish (40) — 3 findings + 1 recommendation; screenshots
2:40–3:00	Dry-run (20) — rehearse the 3-minute story
Hour 4	Presentations & closing — 3 min per group

Hard checkpoint at 1:15: if a role is stuck, hand the partial output to the group and move on. Done-small beats perfect.

Handy snippets

```
import pandas as pd
# Scraper/Cleaner – scrape + clean a practice portal
url = 'https://datasciencebaguio-production.up.railway.app/car_data1' # or car_data2 / car_data3
df = pd.read_html(url)[0]
df = df.drop_duplicates().copy()
for c in df.select_dtypes('number').columns:
    df[c] = df[c].fillna(round(df[c].mean()))
print(df.shape, "rows after cleaning")
```

```
# Analyst – read the Collector's Google Sheet (Share: Anyone with link -> Viewer)
SHEET_ID = 'PASTE_SHEET_ID'
df = pd.read_csv(f'https://docs.google.com/spreadsheets/d/{SHEET_ID}/export?format=csv')
print(df.head())
```

Deliverables (tick to pass)

- ☐ Sheet with collected/scraped data ☐ Cleaned dataset + one correlation
- ☐ EDA dashboard + 3 answered questions ☐ One AI-or-app artifact
- ☐ 3-minute story: what data → what we found → what we'd decide

Scoring rubric (20 points)

Criteria	Points
Data collected / scraped — a real Sheet or scraped table	4
Cleaning — duplicates removed, missing handled, 1 correlation shown	4
Analysis — dashboard + 3 questions answered from their own data	4
AI / app artifact — classifier result, AppSheet screen, or n8n alert	4
Presentation — clear 3-min story, ends in a decision/recommendation	4
Total	20